

Nessus Vulnerability Scan Report

Target: localhost (127.0.0.1)

Tool: Nessus Essentials (Free Edition)

Scan Type: Basic Network Scan

Date: 26 June 2025

Scanned By: Suraj Mishra

Scan Summary

Severity	Count	Examples
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Critical	0	-
High	0	-
Medium	1	SMB Signing Not Required
Informational	6+	SSL Weak Protocols, OS Detection, etc.

Top Vulnerabilities Identified

1. SMB Signing Not Required
- Severity: Medium (CVSS Score: 5.3)

- Description: SMB (Server Message Block) protocol is used for file and printer sharing. If signing is not required, it can be exploited by attackers for man-in-the-middle attacks.

- Fix:

- Enable SMB Signing via Group Policy Editor:

Computer Configuration > Windows Settings > Security Settings > Local Policies > Security Options

-> Microsoft network client: Digitally sign communications (always) -> Enable

-> Microsoft network server: Digitally sign communications (always) -> Enable

2. SSL/TLS Weak Protocols Detected

- Severity: Informational
- Description: The system supports outdated SSL/TLS versions or cipher suites.
- Fix:
 - Use only TLS 1.2 or 1.3.
 - Disable SSL 2.0, SSL 3.0, and TLS 1.0/1.1

3. Netstat Port Scanner Detected Open Ports

- Severity: Informational
- Description: Nessus was able to find open ports using Netstat.
- Ports Found: Common system ports such as 135, 139, 445
- Fix:
 - Close unused ports using Windows Defender Firewall
 - Monitor for suspicious open ports regularly

4. Operating System Information Disclosure

- Severity: Informational
- Description: Nessus was able to detect the OS type and version remotely.
- Fix:
 - Limit exposed services
 - Use firewall rules to block OS fingerprinting tools

Fix Summary Table

Vulnerability	Severity	Fix
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SMB Signing Not Required | Medium | Enable SMB Signing via Group Policy

SSL Weak Protocols | Info | Disable SSL/TLS <1.2, Use TLS 1.3

Netstat Portscanner | Info | Close unused ports via firewall

OS Info Disclosure | Info | Block fingerprinting services

Conclusion

This scan helped identify medium-level and informational vulnerabilities in the local system. The most impactful issue was SMB Signing not required, which can be fixed via local policies. Other issues like SSL/TLS configuration and exposed ports can be resolved by updating protocols and firewall rules.

Report Prepared By:

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Cyber Security Internship Task 3

Date: 26 June 2025